

Minimum Substring

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1 Description

We can scramble a string s to get a string t using the following algorithm:

- If the length of the string is 1, stop.
- If the length of the string is > 1 , do the following:
 - Split the string into two non-empty substrings at a random index, i.e., if the string is s , divide it to x and y where $s = x + y$.
 - Randomly decide to swap the two substrings or to keep them in the same order. i.e., after this step, s may become $s = x + y$ or $s = y + x$.
 - Apply step 1 recursively on each of the two substrings x and y .

Given two strings $s1$ and $s2$ of the same length, return true if $s2$ is a scrambled string of $s1$, otherwise, return

- **Input:**
 $s1 = \text{"great"}$
 $s2 = \text{"rgeat"}$
- **Output:**
 "true"

2 Function/Method Declaration

```
class Solution {
public:
    bool isScramble(std::string s1, std::string s2) {
        /*
            s1 - string pattern
            s2 - string target. Scrambling s1 must follow this target pattern structure
        */
    }
}
```

3 How to solve

$s1$ - data
 $s2$ - target rule

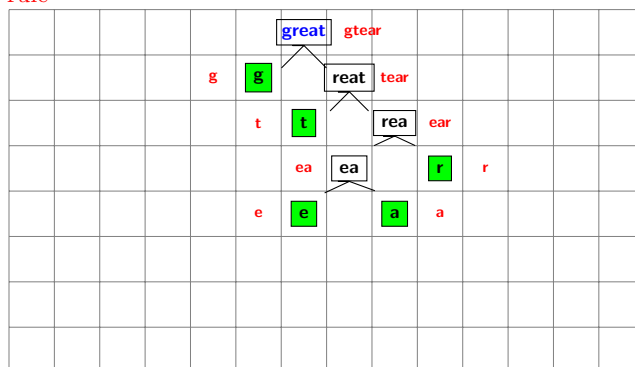


Figure 1: Branch network created by solver.